

15-Week Study Checklist

AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

Study about 1-2 hours a day. Tick each box when done. Do not move on until you finish the CHECKPOINT of each week.

Team names: _____ Start date: _____

Stage	Weeks	Goal
1 Learn the basics	1-3	Python, audio, simple ML
2 Collect dataset	4	Real + fake voices, split data
3 Extract features	5-6	Spectrograms, baseline model
4 Train model	7-9	CNN and pretrained wav2vec2
5 Test and improve	10-11	Honest testing, augmentation
6 Real-time detection	12-13	Live mic + simple app
7 Prevention layer	14	Alert, log, challenge phrase
Finish	15	Report, demo video, slides

Rules that save you: (1) save work often to Google Drive + GitHub; (2) stuck over an hour? search the exact error message or ask; (3) if behind, use only the pretrained model in Stage 4 but never skip testing; (4) split tasks in the team, but everyone finishes Weeks 1-3.

STAGE 1: Learn the basics (Weeks 1-3)

Week 1: Python from zero

Done: _____

- ☐ Day 1: Open Google Colab, run `print("Hello")`, learn variables and data types
- ☐ Day 2: if/else decisions and comparisons
- ☐ Day 3: Loops (for, while)
- ☐ Day 4: Functions (reusable blocks of code)
- ☐ Day 5: Lists and dictionaries
- ☐ Day 6-7: Practice: number-guessing game and simple calculator

☐ **CHECKPOINT: Write a program that takes a list of numbers and prints the average, biggest and smallest.**

Week 2: Data tools and audio basics

Done: _____

- ☐ Day 1: Install libraries with pip; learn NumPy (arrays of numbers)
- ☐ Day 2: Pandas (tables) and reading a CSV file
- ☐ Day 3: Matplotlib for drawing graphs
- ☐ Day 4: What is sound? Waveform, frequency, sampling rate (16,000/sec)
- ☐ Day 5: Load a .wav file with librosa and plot its waveform
- ☐ Day 6: Plot a spectrogram; compare your voice with another recording
- ☐ Day 7: Revise anything confusing

☐ **CHECKPOINT: Record 5 voice clips, load them in Colab, show each waveform and spectrogram.**

Week 3: Machine learning and voice-cloning concepts

Done: _____

- ☐ Day 1-2: What is ML? Supervised learning, features, labels (Google ML Crash Course)
- ☐ Day 3: Overfitting, accuracy, train/validation/test split
- ☐ Day 4: Train a simple scikit-learn classifier (Iris dataset)
- ☐ Day 5: Rough idea of neural networks: layers, weights, training
- ☐ Day 6: How voice cloning works and where clones leave traces
- ☐ Day 7: Read 2-3 blog posts on audio deepfake detection

☐ **CHECKPOINT: Explain to a friend in 5 minutes how a model learns to tell two things apart.**

STAGE 2: Collect the dataset (Week 4)

Week 4: Data collection

Done: _____

- ☐ Day 1-2: Download a small dataset (Fake-or-Real is beginner-friendly); use 1,000-2,000 clips first
- ☐ Day 3: Organize folders real/ and fake/; list files and labels in a Pandas table
- ☐ Day 4: Check the data: clips per class, listen to samples, check durations
- ☐ Day 5: Split into train 70%, validation 15%, test 15% (do NOT touch test set yet)
- ☐ Day 6-7: Optional: generate 20-50 fake clips with a free TTS tool (e.g. Coqui TTS)

☐ **CHECKPOINT: A table with one row per audio file: its label and its split.**

STAGE 3: Extract audio features (Weeks 5-6)

Week 5: Learn and build feature extraction

Done: _____

- ☐ Day 1-2: Learn what MFCCs and mel-spectrograms are
- ☐ Day 3-4: Write a function: audio file in, mel-spectrogram out (librosa)
- ☐ Day 5: Make all clips the same length (e.g. 4 seconds) by trimming/padding
- ☐ Day 6-7: Plot real vs fake spectrograms side by side; look for differences

Week 6: Process everything and set a baseline

Done: _____

- ☐ Day 1-2: Run extraction on the whole dataset; save as .npy files
 - ☐ Day 3: Normalize the features
 - ☐ Day 4-5: Train a simple scikit-learn baseline (Random Forest or SVM) on MFCC averages
 - ☐ Day 6-7: Write down the baseline accuracy; later models must beat it
- ☐ **CHECKPOINT: Saved features for every clip, plus a baseline accuracy number.**

STAGE 4: Train the detection model (Weeks 7-9)

Week 7: Deep learning basics

Done: _____

- ☐ Day 1-3: PyTorch (or Keras): tensors, layers, loss, optimizer, training loop
- ☐ Day 4-5: Train a tiny neural network on a toy dataset
- ☐ Day 6-7: Load your spectrograms into a Dataset and DataLoader

Week 8: Your first CNN

Done: _____

- ☐ Day 1-2: Learn what a CNN is (it reads a spectrogram like an image)
- ☐ Day 3-5: Build and train a small CNN; switch on Colab's free GPU
- ☐ Day 6-7: Plot training vs validation loss and accuracy

Week 9: Pretrained model

Done: _____

- ☐ Day 1-2: Learn what pretrained and fine-tuning mean
 - ☐ Day 3-5: Try wav2vec2 from Hugging Face (fine-tune, or extract features + small classifier)
 - ☐ Day 6-7: Compare with CNN and baseline; save the best model file
- ☐ **CHECKPOINT: A saved model that beats your baseline on the validation set.**

STAGE 5: Test and improve (Weeks 10-11)

Week 10: Honest testing

Done: _____

- ☐ Day 1-2: Run on the untouched test set: accuracy, precision, recall, confusion matrix
- ☐ Day 3: Learn EER (equal error rate) and compute it
- ☐ Day 4-5: Test on fakes from a different generator or dataset (expect a drop)
- ☐ Day 6-7: List where the model fails (noise, phone quality, short clips)

Week 11: Improvement

Done: _____

- ☐ Day 1-2: Data augmentation: background noise, volume and speed changes
- ☐ Day 3: Simulate phone-call quality (lower sampling rate, compression)
- ☐ Day 4-5: Retrain and tune (learning rate, batch size)
- ☐ Day 6-7: Re-test and update your results table

☐ **CHECKPOINT: A results table showing before and after improvement.**

STAGE 6: Real-time detection (Weeks 12-13)

Week 12: Live audio

Done: _____

- ☐ Day 1-2: Install sounddevice; record from the microphone in Python
- ☐ Day 3-4: Cut the live stream into 3-second chunks with small overlap
- ☐ Day 5-6: Run each chunk through your model; print Real/Fake with confidence
- ☐ Day 7: Measure delay (aim for under about 1 second per chunk)

Week 13: Interface

Done: _____

- ☐ Day 1-3: Build a Streamlit or Gradio app: start button, live status, confidence bar
- ☐ Day 4-5: Smooth results by averaging the last 3-5 chunks
- ☐ Day 6-7: Test with friends' voices and AI voices

☐ **CHECKPOINT: A working app that reacts to live speech.**

STAGE 7: Prevention layer (Week 14)

Week 14: Prevention

Done: _____

- ☐ Day 1-2: Add a threshold: if fake score stays high, trigger a visible/audible warning
- ☐ Day 3: Log every alert with timestamp and score to a file
- ☐ Day 4-5: Add a challenge step: caller must repeat a random phrase (defeats pre-recorded clones)
- ☐ Day 6: Optional: email or SMS alert using a free service
- ☐ Day 7: Test the whole flow end to end

☐ **CHECKPOINT: Detect -> alert -> log -> challenge works from start to finish.**

FINISHING (Week 15)

Week 15: Report and presentation

Done: _____

- ☐ Day 1-3: Write the report: problem, method, dataset, results table, limitations
- ☐ Day 4-5: Record a 2-3 minute demo video
- ☐ Day 6-7: Prepare slides and rehearse your presentation

Final check before submission

- ☐ Code runs from start to finish on a fresh Colab session
- ☐ Results table, limitations and future work are written down
- ☐ Demo video recorded; slides ready; every team member can explain every stage